Q1. Consider any first-order difference equation, but with the right-hand side as a function of time. Explain the solution method using a simple example.

Introduction Lag operator such that : $L_{yt}=y_{t-1}$, $L^n yt = y_{t-n}$, $L^{-1}yt = y_{t+1}$.

Starting with a general difference equation :

$$C_1Y_t + C_oY_{t-1} = g(t),$$

y_t is the unknown sequence you want to solve for
g(t) is a known input sequence

Assumption : $g(t)=x_t$; The right-hand side is some given sequence of real values

Normalisation : We divide by C_1 (the coefficient of Y_t, so that we get only Y_t in the equation)

$$y_t+c_0/c_1(y_{t-1})=1/c_1(x_t)$$

Instead of writing $1/c_1(x_t)$ every time, we introduce a new sequence

$$X_t \equiv 1/c_1(x_t)$$

This is a definition, It means: from now on, whenever we say X_t , we mean $1/c_1(x_t)$
So the equation becomes,

$$y_t+by_{t-1}=X_t$$

$b = c_0/c_1$
Putting in the lag operator, replacing y_t+by_{t-1} with L_{yt}

$$y_t+b(L_{yt})=X_t$$

Factoring out Y_t
$(1+bL)y_t=X_t$

$$Y_t = (1+bL)^{-1}X_t \text{ (this will be the particular solution)}$$

Homogeneous solution
set $X_t=0$

$$y_t^{(h)}+by_{t-1}^{(h)}=0 \Rightarrow y_t^{(h)}= -by_{t-1}^{(h)}$$

$$y_t^{(h)}=C(-b)^t \text{ (C is an arbitrary constant)}$$

Particular solution

$$Y_t = (1+bL)^{-1}X_t$$

If we let $\alpha = -b$ and apply $(1- \alpha L)^{-1} = 1 + \alpha L + \alpha^2L^2 + ... = \Sigma^a_{(i=0)} (\alpha^iL^i)$
$\bar{y}t = {}_{(i=0)}^{\infty}\sum(-b)^iL^iX_t= {}_{(i=0)}^{\infty}\sum(-b)^iX_{t-i}$ satisfies the non-homogeneous difference-equation, hence it is a particular solution.

Expanding the general solution:
One convenient particular solution (when it converges) is the infinite convolution:

$$y_t^{(p)} = \sum_{\{i=1\}}\infty (-b)^i X_{\{t-i\}}$$

which is valid when the series converges ($|b|<1$)

So, General solution,

The **general solution** is homogeneous + particular:

$$y_t = C(-b)^t + \sum_{(i=0)}^{\infty}(-b)^i X_{t-i} \qquad \textbf{(valid if } |b|<1\textbf{)}$$

The first term $C(-b)^t$ is the transient part, determined by the initial condition.
The second term is the forced response, determined by the input X_t.

If $|b|>1$
The above series diverges. Then we use the alternative expansion:
$(1-\alpha L)^{-1}=-1/\alpha L(1-1/\alpha L1)^{-1}$
$(1 - 1/(\alpha L))^{-1} = 1 + (\alpha L)^{-1} + (\alpha L)^{-2} + \ldots$
Multiply
$(1 - \alpha L)^{\{-1\}} = -1/(\alpha L)(1 + 1/(\alpha L) + 1/(\alpha L)^2 + \ldots) = -1/\alpha\, L^{\{-1\}} - 1/\alpha^2\, L^{\{-2\}} - 1/\alpha^3\, L^{\{-3\}} - \ldots$
Rewrite as a series
$(1 - \alpha L)^{\{-1\}} = -\sum$ (from i=1 to ∞) $(1/\alpha^i)\, L^{\{-i\}}$

Substitute $\alpha = -b$
$(1+bL)^{-1}=(1-\alpha L)^{-1}$, $\alpha=-b$.
$(1+bL)^{-1}= -\sum(\infty_{i=1})\,(1/(-b)^i)L^{-i}$

Apply to X_t
$y_t^{(p)}=(1+bL)^{-1}X_t= -\sum(_{i=1}\infty)(1/(-b)^i)(L^{-i})X_t$

But $(L^{-i})X_t = X_{t+i}$

So

$y_t^{(p)} = -\sum\,(_{i=1}\infty)(1/(-b)^i)X_{t+i}$
Then, **General solution**
Combine homogeneous and particular
$Y_t = C(-b)^t - \sum(_{i=1}\infty)(1/(-b)^i)X_{t+i}$ **(valid if $|b|>1$)**

Example 1: when $|b|<1$ (stable)
Consider : $y_t+½(y_{t-1}) = t$

- Here $b = ½$; $|b|<1$.
- The homogeneous solution is
  $y_t^{(h)}=C(-1/2)^t$
- Particular solution uses the past expansion:
  $y_t^{(p)} = \sum(_{i=0}^{\infty})(-1/2)^i(t-i)$; $X_t=t$.
- General solution:
  $y_t=C(-1/2)^t+\sum(_{i=0}^{\infty})(-1/2)^i(t-i)$

  (The weights $(-1/2)^i$ decay to zero. Expressed in terms of past inputs, which is valid because the weights $(-1/2)^i$ decay to zero.) The particular solution is a weighted average of past inputs X_{t-i}. Today's outcome is explained by past demand shocks, policy actions, or external forces, but their effect weakens the further back in time they are. Eg, adaptive consumption, demand with lags.

Example 2: when $|b|>1$ (unstable)

Consider : $y_t+ 2(y_{t-1}) = t$

- Here $b=2$; $|b|>1$
- The homogeneous solution is
  $y_t^{(h)}=C(-2)^t$
- Particular solution uses the future expansion:
  $y_t^{(p)}= -\sum(_{i=1}^{\infty})1/(-2)^i(t+i)$
- General solution:
  $y_t = C(-2)^t - \sum(_{i=1}^{\infty})(1/(-2)^i)(t+i)$

Expressed in terms of future inputs X_{t+i}, which is the alternative expansion valid when $|b|>1$. This is an *unstable system* if left on its own. Interpretation: when $|b|>1$, you can't explain today's outcome only from the past. It must be pinned down by forward-looking information about future inputs. Eg, Asset pricing, forward-looking inflation expectations.

Intuition of cases:

Stable Systems ($|b|<1$): Think of consumer spending habits. If people adjust their consumption based on past income, but with some moderation ($b<1$), temporary income shocks gradually fade. Today's consumption depends on a weighted average of all past income changes, with recent changes mattering more.

Unstable Systems ($|b|>1$): Consider asset pricing where expectations matter more than fundamentals. Stock prices can't be explained just by past dividends; they must be "pinned down" by expectations of future performance. This forward-looking behavior could prevent explosive bubbles in rational markets.

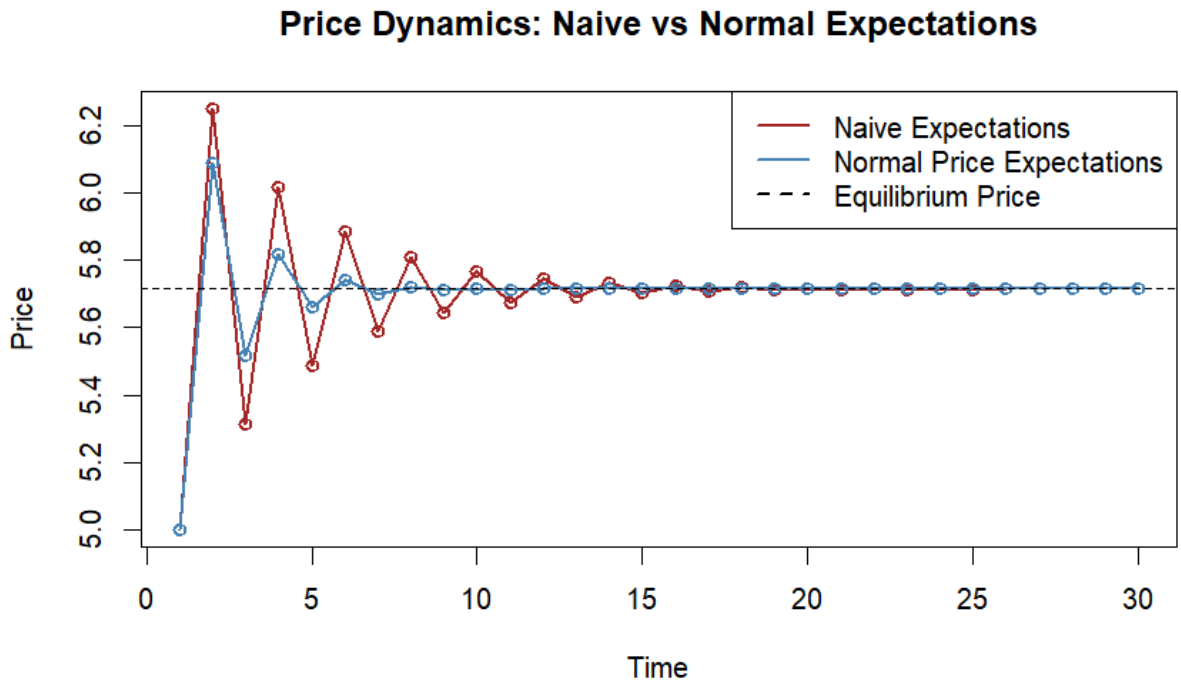
Economic context (real world) :

Macroeconomic Policy: When $|b|<1$ (stable case), monetary policy shocks have diminishing effects over time. Central banks can use interest rate changes knowing their impact will eventually fade, allowing for predictable policy outcomes.

Business Cycles: The transient component $C(-b)^t$ represents temporary deviations from long-run trends, while the particular solution captures how the economy responds to ongoing structural changes.

Investment Dynamics: In capital accumulation models, $|b|>1$ suggests explosive growth requiring forward-looking constraints (like budget limits) to remain realistic.

Q2. Consider the following demand supply model with the usual assumptions for the parameters. Use two different expectations formation functions (Naive, normal price and adaptive) and plot them in the same figure.



The figure shows how prices adjust under two expectation rules. With naive expectations (red line), producers assume next period's price will equal the last one, which generates oscillations around the equilibrium that die out only slowly. Under normal price expectations (blue line), producers form expectations as a weighted average of the equilibrium price and the last observed price, so prices converge more quickly to the equilibrium (dashed green line). The speed of adjustment depends on the parameter ccc: when c=0 the rule reduces to naive expectations, while higher values of ccc put more weight on the equilibrium price and speed up convergence.

<u>Effect of the Expectation Formation Parameter c on Stability</u>

In the generalised cobweb model with normal price expectations, the expected price is defined as

$pt^E=(1-c)pt-1+ cp_N$; $0 \leq c \leq 1$

where pN is the normal (long-run) price and c is the speed of adjustment.

- When c=0, the model reduces to the naive expectations case: suppliers simply use the last observed price, and stability depends only on the relative slopes of supply and demand.

Convergence, if it occurs, is slow.

- When $c=1$, expectations jump directly to the equilibrium price, so the system converges immediately — the most stable case possible.

- For values $0<c<1$, the parameter c acts like a stabiliser: it scales down the feedback term in the price difference equation by a factor of $(1-c)$. The higher c is, the smaller this multiplier becomes, and the faster deviations from equilibrium die out. A lower ccc means weaker expectations, slower adjustment, and more prolonged oscillations.

Formally, the stability condition becomes

$$|s1/d1(1-c)|<1$$

where s1 is the slope of the supply curve and d1 the slope of demand. Compared to the original cobweb model, the extra factor $(1-c)$ widens the range of parameter values for which the system is stable. This means that introducing normal price expectations can stabilize a market that would otherwise diverge under naive expectations.

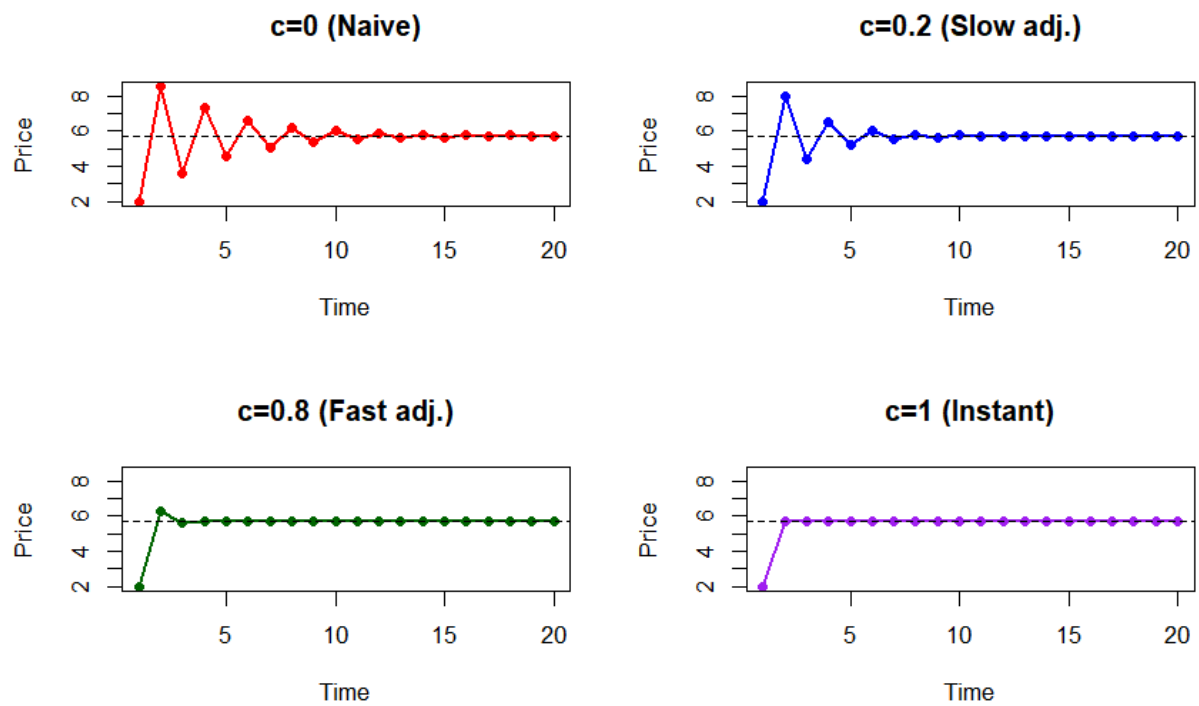
In summary:

- Increasing c speeds up convergence and enhances stability.

- $c=0$ collapses back to the naive model.

- $c=1$ gives instantaneous convergence.

- For intermediate values, stability is retained as long as the scaled slope ratio satisfies the condition above, with higher c leading to quicker damping of oscillations.



<u>Discuss the result vis-a-vis the case of naive expectations.</u>

Under naïve expectations (c=0), suppliers assume that the next period's price will simply equal the last observed price. This collapses the generalised cobweb model into the original one, where equilibrium stability depends solely on the relative slopes of demand and supply: stability holds if $|s1/d1|<1$. In this case, even when the market does converge, the process is typically slow and oscillatory, since expectations always lag behind actual market movements. Agents are essentially "chasing" the price rather than anticipating it, which can amplify cycles when conditions are unfavorable.

By contrast, under normal price expectations ($0<c\leq1$) , suppliers anchor their forecasts to the long-run equilibrium price pN. The adjustment parameter c measures the speed of convergence toward that benchmark:

- When c=0, the system collapses back to the naïve case.

- When $0<c<1$, the stability condition becomes $|s1/d1(1-c)|<1$. The factor $(1-c)$ dampens the effective feedback loop, broadening the range of slope ratios for which the system remains stable. As a result, convergence is not only more likely but also faster, with the speed increasing in c (e.g., convergence at c=0.8 is much quicker than at c=0.2). Making $c = 0$ eliminates the (1-c), keeping only the naive expectation $|s1/d1|$.

- When c=1, expectations jump immediately to the equilibrium price, so the market converges in a single step.

$$d_0 + d_1 p_t = s_0 + s_1 p_t^e \quad \Rightarrow \quad p_t = \frac{s_0 - d_0}{d_1} + \frac{s_1}{d_1} p_t^e.$$

Define deviations $x_t = p_t - p_N$, and note p_N satisfies the same identity with $p_t^e = p_N$, therefore

$$x_t = \frac{s_1}{d_1}(p_t^e - p_N).$$

For normal-price expectations $p_t^e = (1-c)p_{t-1} + cp_N$, so $p_t^e - p_N = (1-c)(p_{t-1} - p_N) = (1-c)x_{t-1}$. Hence

$$x_t = \frac{s_1}{d_1}(1-c)\, x_{t-1},$$

so stability requires $\left|\frac{s_1}{d_1}(1-c)\right| < 1.$

In short, compared with the naïve framework, normal price expectations embed a stabilising mechanism: agents anticipate the pull toward equilibrium rather than just extrapolating from the past. This makes markets more resilient to shocks and ensures that deviations are corrected more quickly. Indeed, cases that would diverge under naïve expectations can be stabilised once forward-looking elements are introduced.

Economic context

Naive Expectations (c=0): Represents markets with poor information flow or inexperienced traders. Think of emerging markets where participants lack sophisticated forecasting tools and simply extrapolate from recent observations. This creates persistent volatility as everyone "chases" price movements.

Normal Price Expectations (c>0): Reflects markets with better fundamental analysis. Professional commodity traders, for instance, don't just look at yesterday's wheat prices; they consider long-term supply/demand fundamentals, weather patterns, and policy changes. The parameter 'c' measures market sophistication.

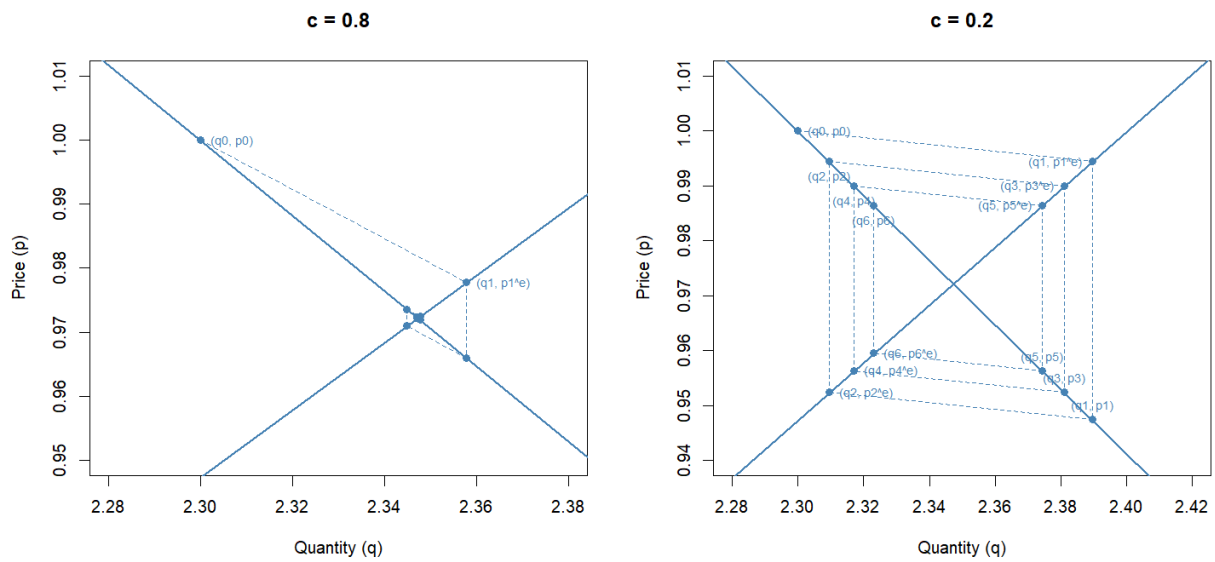
Economic implications

Central Banking: When forming inflation expectations, if c is low (naive expectations), monetary policy needs to be more aggressive to anchor expectations. If c is high (forward-looking), even announcements can stabilize prices without actual intervention.

Agricultural Markets: Government price support programs work better when c is high because farmers quickly incorporate policy announcements into their planting decisions.

Bond Markets: Government bond yields reflect expectations formation. In stable economies (high c), yields quickly incorporate policy announcements. In unstable economies (low c), yields remain volatile as investors rely more on recent price movements.

Q3. Consider the demand-supply model in Chapter 3 of Raghavendra and Piiroinen book (An Introduction to Economic Dynamics). Reproduce the two cobweb plots (Figure 3.2) 1 in page 43. Explain the dynamics of the model through these plots. Compare the Figure 3.2 with Figure 2.5 (chap ter 2) in page 28 and explain the main difference between the two, and what is the mechanism that drives this difference



The two cobweb plots illustrate how expectations shape the adjustment of prices and quantities toward equilibrium.

1. Faster convergence (c = 0.8)

    - Here, suppliers strongly believe that the market price will quickly move toward the "normal" or equilibrium price.
    - This confidence means they adjust their expectations aggressively: the expected price jumps closer to equilibrium in each step.
    - As a result, the actual price and quantity also converge quickly. The path you see is very short.
    - Economically, this reflects a situation where market participants trust that deviations won't last long. For instance, if there's a sudden shock in crop prices, farmers believe the market will stabilize quickly, so they don't overreact in their supply decisions.

2. Slower convergence (c = 0.2)

    - With a lower adjustment speed, suppliers revise their expectations only slightly toward the normal price in each period.
    - The cobweb therefore takes longer to spiral into equilibrium, producing multiple distinct steps where you can clearly trace (qt,pt) and (qt,pt^e).
    - Economically, this captures a cautious supplier, they don't fully trust that today's deviation will vanish soon, so they update slowly. This kind of behavior can keep the market in disequilibrium for longer.

A higher c essentially "compresses time": expectations jump more, convergence is sharper, and disequilibrium periods are short-lived. A lower c "stretches time": convergence still happens, but disequilibrium dynamics (oscillations or drifts) are visible for longer. Think of commodity markets: when farmers expect prices to revert quickly (high c), speculation dampens, and convergence to equilibrium is fast. But if expectations adjust slowly (low c), market participants may prolong mismatches between supply and demand, which could translate into more persistent price fluctuations. In modern financial markets, this is analogous to whether traders quickly incorporate "fundamental values" into pricing (high c) or whether they wait for confirmation (low c).

1. Figure 2.5 (Original Cobweb Model)

- In the basic cobweb model, supply at time *t* depends on the previous period's price (p_{t-1}).
- This lagged adjustment creates the familiar spiral/haggling pattern of the cobweb:
    - If the demand curve is steeper than supply, prices converge to equilibrium in oscillations.
    - If supply is steeper, oscillations diverge and the system becomes unstable.

- Intuitively, sellers "haggle" based on what price they saw last period, not on what they think will happen.

2. Figure 3.2 (Expectations in the Cobweb Model)

- Here, supply depends on the expected price, not just the last observed price.
- Specifically, suppliers form expectations about a normal price (p^N, the equilibrium price), and the adjustment speed is controlled by parameter *c*.

    - High c (0.8), suppliers strongly expect quick convergence to the normal price, cobweb lines are slanted toward equilibrium, system converges faster.

- Low c (0.2), weaker expectation, suppliers rely more on lagged adjustment, cobweb looks more like the classic oscillatory case, with slower convergence.

The Main Difference :

No expectations:
Dynamics are driven by mechanical lagged supply responses to last period's price. Adjustment is slow, oscillatory, and can diverge.

With expectations:
Dynamics are driven by forward-looking expectations. The "slant" in the cobweb comes from the fact that suppliers don't just copy last period's price but anticipate movement toward the equilibrium.

- This expectation acts like a shortcut in the haggling process: instead of waiting to see the actual price adjustment, suppliers adjust in advance toward the normal price.
- Mechanism: the *expectation parameter (c)* scales the adjustment. A higher *c* reduces the amplitude of oscillations and accelerates convergence.

Mechanism Driving the Difference

- In no expectations, Haggling & lagging drives, market participants only react to yesterday's price, leading to slower, oscillatory paths.

- In Normal prices with expectations: Expectation-based slanting drives, participants incorporate beliefs about the equilibrium, tilting the cobweb toward faster convergence.

- The key mechanism: expectations replace pure lag dependence. Instead of being backward-looking, the system becomes forward-looking, and this makes the dynamics more stable and faster (depending on *c*).

The main difference between Fig. 2.5 and Fig. 3.2 is that expectations create a slant in the cobweb paths, accelerating or slowing convergence depending on how strongly agents expect a return to equilibrium. The mechanism is the shift from lagged "haggling" behavior (backward-looking supply) to expectation-driven adjustment (forward-looking supply). In Fig. 2.5, expectations are fully backward-looking (naive), so the cobweb depends only on lagged

price. In Fig. 3.2, expectations are partly anchored on the normal price (through *c*), which pulls the system toward equilibrium more quickly when *c* is larger. Therefore, The fundamental difference between Figure 2.5 (naive) and Figure 3.2 (expectations) lies in **information processing**.

Further, Let $\alpha = s1/d1(1-c)$ stability requires $|s1/d1|<1$

Monotone convergence:
If $0<\alpha<1$, then successive deviations shrink without changing sign. Prices approach the normal price smoothly.

- Oscillatory convergence:
  If $-1<\alpha<0$, then successive deviations alternate in sign while shrinking in magnitude. This generates the familiar "zig-zag" cobweb pattern converging to equilibrium.

- Divergence:
  If $|\alpha|>1$, the process is unstable, deviations grow over time.

In economic terms, whether convergence is monotone or oscillatory depends on the sign of d1 and s1. With the usual case of $d1<0$(downward-sloping demand) and $s1>0$(upward-sloping supply), we have $s1/d1<0$. Therefore, the adjustment factor α is typically negative, so convergence (when it occurs) is oscillatory rather than monotone. A larger value of c (closer to 1) reduces $|\alpha|$, making convergence faster.

Economic context

Production Lags: The cobweb model can show agricultural cycles. Farmers must decide how much to plant based on expected prices, but can't adjust supply immediately when prices change. This creates the fundamental lag that drives cobweb dynamics.

Fast vs Slow Learning (c=0.8 vs c=0.2):

- High c (0.8): Modern precision agriculture with futures markets, weather derivatives, and sophisticated forecasting. Farmers quickly incorporate long-term price signals, leading to stable markets.
- Low c (0.2): Traditional subsistence farming with limited market information. Farmers rely mainly on recent price experience, creating persistent boom-bust cycles.

Stock Markets: Generally high-c systems due to professional analysis, but can switch to low-c during panic periods.

Commodity Markets: Agricultural commodities often low-c due to production lags; financial commodities (gold, oil futures) often high-c due to speculation.

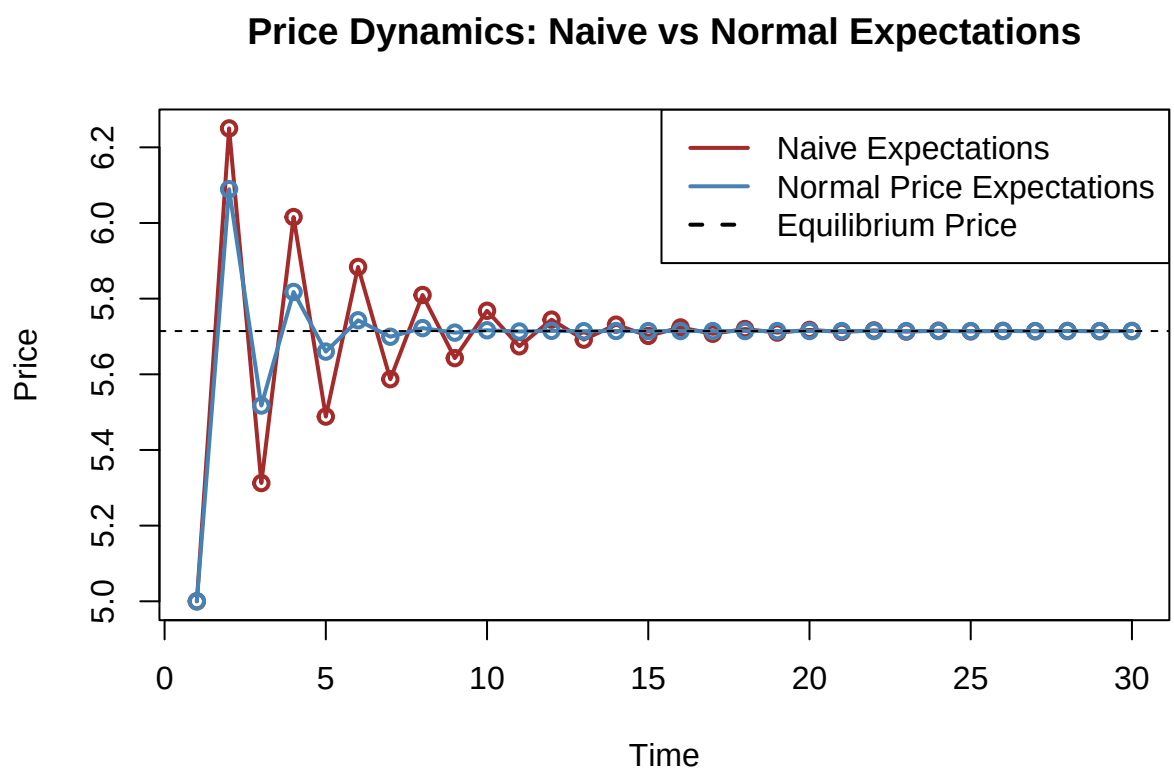
DDD_Assignment 2

Michelle

2025-09-12

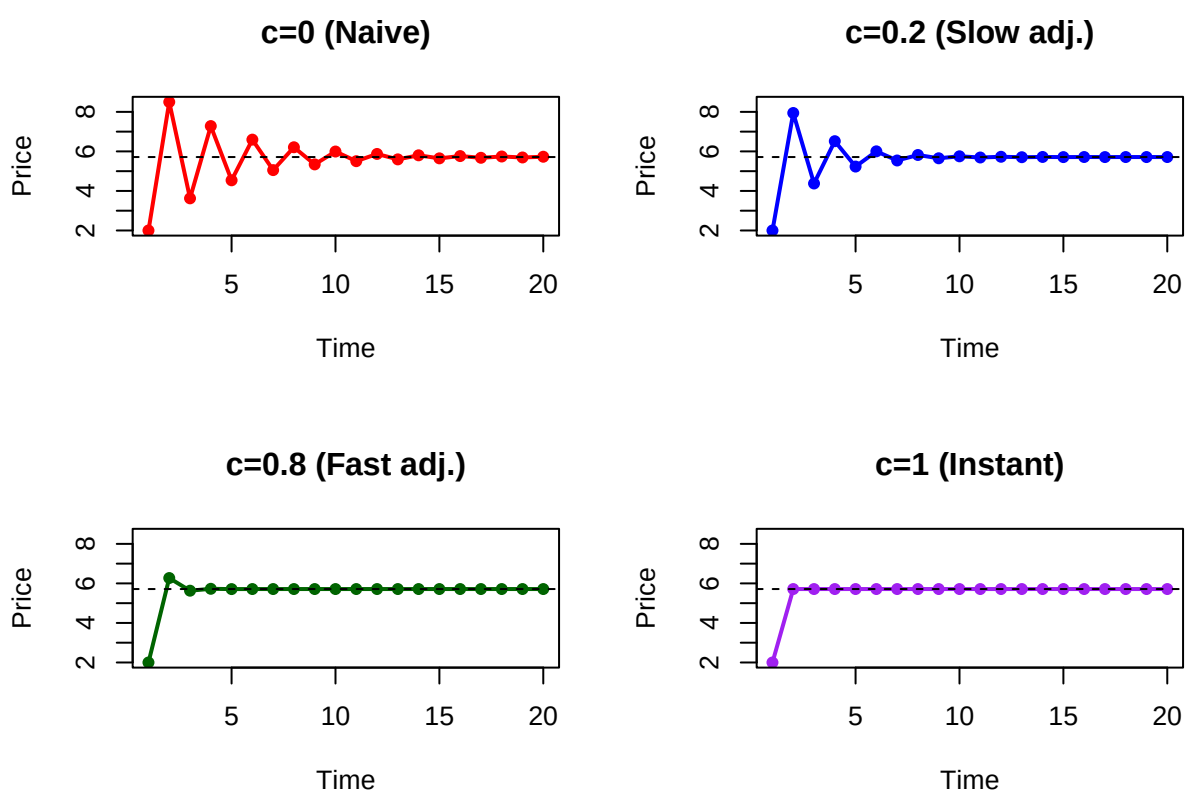
```r
rm(list = ls())
# Demand: Qd = a + b*pt
# Supply: Qs = a1 + b1*pt^E

# Parameters
a  <- 50     # demand intercept
b  <- -4     # demand slope (<0)
a1 <- 10     # supply intercept
b1 <- 3      # supply slope (>0)

# Equilibrium ("normal") price
pN <- (a1 - a) / (b - b1)

# Expectation formation parameter for "normal price expectations"
c <- 0.3   # 0 < c <= 1

# Time horizon
T <- 30
p_naive  <- numeric(T)
p_normal <- numeric(T)

# Initial price
p_naive[1]  <- 5
p_normal[1] <- 5

# Dynamics
for (t in 2:T) {
  # Naive expectations
  pE_naive <- p_naive[t-1]
  p_naive[t] <- (a1 - a)/b + (b1/b) * pE_naive

  # Normal price expectations
  pE_normal <- c * pN + (1-c) * p_normal[t-1]
  p_normal[t] <- (a1 - a)/b + (b1/b) * pE_normal
}

# Plot
plot(1:T, p_naive, type="o", col="brown", lwd=2, ylim=range(c(p_naive, p_normal, pN)),
     xlab="Time", ylab="Price", main="Price Dynamics: Naive vs Normal Expectations")
lines(1:T, p_normal, type="o", col="steelblue", lwd=2)
abline(h=pN, lty=2)
legend("topright", legend=c("Naive Expectations", "Normal Price Expectations", "Equilibrium Price"),
```

```
       col=c("brown", "steelblue", "black"), lty=c(1,1,2), lwd=2)
```

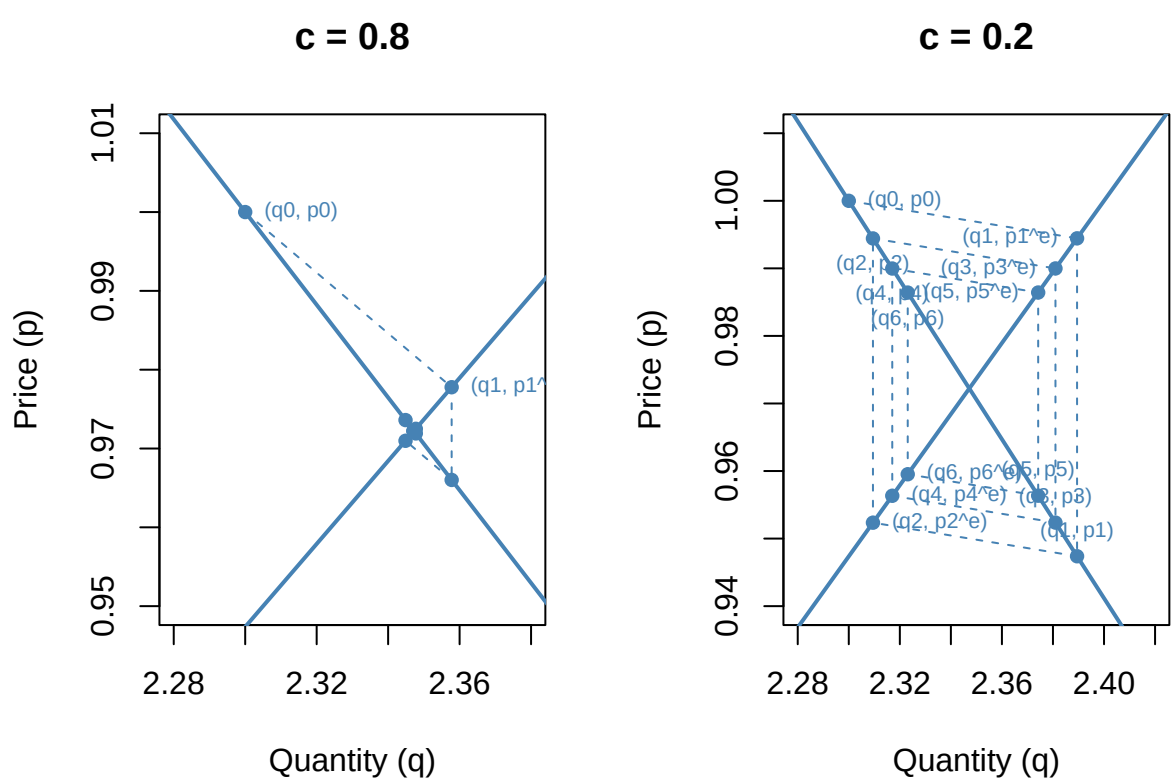



```
# Parameters
a  <- 50
b  <- -4
a1 <- 10
b1 <- 3

# Equilibrium price
pN <- (a1 - a) / (b - b1)

# Time horizon
T <- 20
c_vals <- c(0, 0.2, 0.8, 1)
labels <- c("c=0 (Naive)", "c=0.2 (Slow adj.)",
            "c=0.8 (Fast adj.)", "c=1 (Instant)")
colors <- c("red", "blue", "darkgreen", "purple")

# Store results in a list
prices <- vector("list", length(c_vals))
names(prices) <- labels

for (j in seq_along(c_vals)) {
  c <- c_vals[j]
  p <- numeric(T)
```

```r
  p[1] <- 2  # initial price
  for (t in 2:T) {
    pE <- c * pN + (1 - c) * p[t-1]
    p[t] <- (a1 - a)/b + (b1/b) * pE
  }
  prices[[j]] <- p
}

# Plot 2x2 facets
par(mfrow=c(2,2))  # 2 rows, 2 columns

for (j in seq_along(c_vals)) {
  plot(1:T, prices[[j]], type="o", col=colors[j], lwd=2, pch=16,
       ylim=range(unlist(prices), pN),
       xlab="Time", ylab="Price", main=labels[j])
  abline(h=pN, lty=2)
}
```




```r
rm(list = ls())

# Parameters
d0 <- 4
d1 <- -1.7
s0 <- 0.5
s1 <- 1.9
```

```r
p0 <- 1
n  <- 6   # steps

# Function to simulate cobweb
simulate_cobweb <- function(c, d0, d1, s0, s1, p0, n) {
  p  <- numeric(n + 1)
  pe <- numeric(n + 1)
  D  <- numeric(n + 1)
  S  <- numeric(n + 1)

  pN <- (s0 - d0) / (d1 - s1)   # equilibrium price

  p[1]  <- p0
  pe[1] <- p0
  D[1]  <- d0 + d1 * p[1]
  S[1]  <- s0 + s1 * pe[1]

  for (k in 1:n) {
    pe[k + 1] <- p[k] + c * (pN - p[k])
    p[k + 1]  <- (s1 / d1) * pe[k + 1] + (s0 - d0)/d1
    D[k + 1]  <- d0 + d1 * p[k + 1]
    S[k + 1]  <- s0 + s1 * pe[k + 1]
  }

  list(p = p, pe = pe, D = D, S = S)
}

# Simulations
res_fast <- simulate_cobweb(c = 0.8, d0, d1, s0, s1, p0, n)
res_slow <- simulate_cobweb(c = 0.2, d0, d1, s0, s1, p0, n)

# Plotting function
plot_cobweb <- function(res, cval, xlim, ylim, label_mode=c("full","minimal")) {
  label_mode <- match.arg(label_mode)

  plot(NA, xlim = xlim, ylim = ylim,
       xlab = "Quantity (q)", ylab = "Price (p)",
       main = paste("c =", cval))

  # Demand curve
  lines(c(0, d0), c(-d0/d1, 0), col = "steelblue", lwd=2)
  # Supply curve
  lines(c(s0, d0), c(0, (d0 - s0)/s1), col = "steelblue", lwd=2)

  for (k in 2:(length(res$p))) {
    # vertical step (dashed)
    segments(res$D[k-1], res$p[k-1], res$S[k], res$pe[k],
             col="steelblue", lty=2)
    points(res$S[k], res$pe[k], pch=16, col="steelblue")

    # horizontal step (dashed)
    segments(res$S[k], res$pe[k], res$D[k], res$p[k],
             col="steelblue", lty=2)
```

```r
    points(res$D[k], res$p[k], pch=16, col="steelblue")

    # Labels depending on mode
    if (label_mode == "full") {
      text(res$S[k], res$pe[k],
           paste0("(q", k-1, ", p", k-1, "^e)"),
           pos = ifelse(k %% 2 == 0, 2, 4), cex=0.7, col="steelblue")
      text(res$D[k], res$p[k],
           paste0("(q", k-1, ", p", k-1, ")"),
           pos = ifelse(k %% 2 == 0, 3, 1), cex=0.7, col="steelblue")
    } else if (label_mode == "minimal" && k==2) {
      # Only show label for first expectation point
      text(res$S[2], res$pe[2], "(q1, p1^e)", pos=4, cex=0.7, col="steelblue")
    }
  }

  # Always label initial point
  points(res$D[1], res$p[1], pch=16, col="steelblue")
  text(res$D[1], res$p[1], "(q0, p0)", pos=4, cex=0.7, col="steelblue")
}

# Plot both
par(mfrow = c(1, 2))
plot_cobweb(res_fast, 0.8, xlim=c(2.28, 2.38), ylim=c(0.95, 1.01), label_mode="minimal")
plot_cobweb(res_slow, 0.2, xlim=c(2.28, 2.42), ylim=c(0.94, 1.01), label_mode="full")
```